Fiscal Year:	FY 2010	**Task Last Updated:**	FY 09/10/2010
PI Name:	Mitchell, Cary Ph.D.		
Project Title:	(NSCORT) for Advanced Life Support at Purdue/Howard/AAMU - Integrated Systems Group		
Division Name:	Human Research		
Program/Discipline:	ADVANCED HUMAN SUPPORT TECHNOLOGIES		
Program/Discipline--Element/Subdiscipline:	ADVANCED HUMAN SUPPORT TECHNOLOGIES--Advanced life support		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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City:	West Lafayette	**State:**	IN
Zip Code:	47907-2010	**Congressional District:**	4
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	NSCORT
Start Date:	12/01/2002	**End Date:**	11/30/2009
No. of Post Docs:	0	**No. of PhD Degrees:**	1
No. of PhD Candidates:	1	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA JSC
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:	NOTE: Received NCE to 11/30/2009 per PI (1/09) NOTE: Received NCE to 11/30/2008 per PI (11/07)		
Key Personnel Changes/Previous PI:	0		
COI Name (Institution):	Gardner, Guy (No Longer at Purdue) Pekny, Joe (Purdue University) Chiu, George (Purdue University) Yao, Bin (Purdue University) Yih, Yuehwern (Purdue University) Orcun, Seza (Purdue University)		
Grant/Contract No.:			
Performance Goal No.:			
Performance Goal Text:			

Task Description:	Throughout the life of ALS NSCORT, the systems group researchers matured from learning and exploratory mode, which can be mainly characterized by proof-of-concept studies to build a viable toolbox of systems analysis methodologies suitable for ALS analysis, e.g. SimOpt (SIMulation based OPTimization), Markov Theory and Model Predictive Control, towards well structured and coordinated efforts to improve their readiness level for repeated use in answering questions that matters to NASA and ALS community at large. As systems group was undergoing through its voyage within ALS NSCORT, the systems group's analysis and modeling scope has been expanded to all ALS technologies and to adopt lifecycle management view to study 15-20 year Mars Base ALS system evolution which recognizes the fact that Mars Base ALS system will not stay as it was deployed at the very first mission. Unfortunately, our plans and resources had to be revised per early termination notice as NASA was undergoing through major re-organization and re-prioritization to maximize the return on NASA's investment in ALS NSCORT by focusing on completing the matured on going tasks and retiring the new initiatives such as, the proof-of-concept study that has been initiated in collaboration with Envision Center at Purdue University on visualizing the evolution of Mars Base and its ALS system. Throughout 4 years of ALS NSCORT program, systems group has trained 6 graduate students, engaged 11 researchers (3 Post-Docs, 4 original PIs, 4 additional researchers) with ALS NSCORT sponsored systems related projects, contributed to SIMA Lunar Challenge and BVAD, reached out to over 400 undergraduate students by challenging them with ALS driven projects in ChE 450 (Chemical Engineering Design Course) and voluntary undergraduate research projects (Rebecca Alway-Cooper of Chemical Engineering), supported other projects and programs within ALS NSCORT by giving invited lectures at the McKenzie Career Center to Mr. Martin's high school class, editing a "baby BVAD" intended to be used in the project "Equivalent Systems Mass Analysis of Plant Growth" with high school students lead by outreach group of ALS-NSCORT, leading and supporting center-wide project management, roadmap development, process map revisions and ESM evaluation of ALS-NSCORT technologies initiatives and activities. ALS NSCORT researchers on the average gave 3 TIM/SIMA telecons per year, generated 18 conference papers (9 ICES, 5 Habitation, 2 AICHE, 3 other), 1 peer reviewed journal paper. We expect that ALS NSCORT scholarly products will continue to emerge in the next couple of years at various journals (there are 6 papers in progress in our pipeline). We, in collaboration with ALS NSCORT management, have further initiated activities: 1) to archive knowledge generated throughout our ALS NSCORT program within the Purdue University Libraries, 2) to reflect our experience through a lessons learned white paper which will be ready for dissemination in couple of months after the early termination of ALS NSCORT. We believe that these initiatives will enable us to respond to large scale ALS research program opportunities rapidly as they emerge. Seza Orcun Integrated Systems Group – Focus Area Lead
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The overall goal of this project is to develop methodologies to ensure crewmembers survive while on space missions. Development of computer models for control of complex processes in life support in a controlled, closed-loop environment.
Task Progress:	Control Strategy for Dynamic Sustainable Systems A simulation is developed to assist decision-making in choosing control policies in a closed environment. This closed environment consists of a simplified representation of the water subsystem based on the documentation by the National Aeronautics and Space Administration (NASA). The water requirement (consumption) and generation are based on the crewmembers' demographics, activity schedules as well as intensity of each activity. The overall goal of this project is to develop methodologies to ensure crewmembers survive while on space missions. The health of this system is the result of the intricate interaction between the crewmembers as well as the water subsystem. The primary measure of the health of the system is the "state" representation of the system. This representation is a conglomeration of various hourly assessments of different aspects of the system. These aspects include the deterministic as well as the stochastic nature of the system. Due to the potentially vast number of possible states in the entire state space, state reduction techniques as well as assessments for infeasible states are applied for state reduction. A reward system is designed for allocating reward/penalty values to system states that satisfy predetermined conditions. Due to the strong stochastic characteristics and the "memoryless" property of the system, the transition of system state can be modeled as a Markov process. Due to the incomplete knowledge of the new system (that is not yet in existence) and its dynamics, process-control strategies need to be developed in such a way that they are comprehensive enough to manage the system given any kind of system behavior, anticipated or otherwise. However, before strategies can be developed, system states need to be captured. In order to explore the possible system states, a Controlled Random Walk (CRW) algorithm is developed to visit more system states in less amount of time, compared to traditional techniques such as the Pure Random Walk and the Round Robin Walk. Transition probabilities are obtained through the simulation of the Markovian model. Capturing "true" state transition probabilities has been a challenge to researchers. In this research, a methodology is developed to minimize the number of data points (or simulation hours) while obtaining the transition probabilities that are closest to the system's steady state probabilities by observing the percentage increase of states visits as well as probabilities convergence based on discrete consecutive time periods. The Kullback-Leibler Divergence distance is used as the foundation of the measurement. It is then modified in order to be used in this research study. A reference index is also defined based on this measure to study the "distance" between each probability distribution from the index. These probabilities, together with the reward values are formulated into appropriate mathematical representation and are solved by applying the Policy Iteration (PI) algorithm. The outcome from PI is an indication of the "best" policy to be applied when the system is in a given state. Various scenarios with different system design, magnitude of stochastic properties and system modules such as the addition of a crop subsystem can be studied to evaluate the benefits of applying this methodology.

	The simulation was done in three stages: Data Collection/State space exploration, System Evaluation with Dynamic Policy and System Evaluation with Fixed Policy. The Data Collection stage was performed through 800,000 ticks (simulation hours) of simulation time. Every set of 100,000 ticks consists of 5 sets of 20,000 ticks, each with a re-initialized starting condition. This is done to prevent the system from being "stuck" in a state or a loop of states. The proposed Controlled Random Walk technique is used during the data collection. Best policies are obtained through every set of 100,000 ticks, yielding 8 sets of best policies. Every set of best policies are applied to a test system for evaluation. Results show that outcomes from applying these best policies are statistically better than using fixed policies, or using random policies. As mentioned above, comprehensive process-control strategies are to be developed for various system states. The system might visit certain states that have not been visited before during data collection. These states will not have a corresponding "best" policy as they are "new" to the control system. In order to counter that problem, machine-learning technique is used. The See 5.0 algorithm (previously ID3) is used to generate a complete decision tree based on every set of best policies obtained from the PI. With the outcome from the See 5.0 algorithm, an action is available for use in the system given any state that the system visits. The outcomes of See 5.0 algorithm are evaluated against the outcomes from the PI using an independent test system. Results from non-parametric tests show that there is no statistical difference between the two, indicating that while policies obtained from See 5.0 perform as good as optimal policies from PI, See 5.0 also provides a complete decision tree, covering the entire state space.
Bibliography Type:	Description: (Last Updated: 04/16/2020)
Abstracts for Journals and Proceedings	Chiam TC. "A rule-based system is constructed based on a Markov Decision Model to determine the selection of dynamic policies for a NASA Advanced Life-Support System." INFORMS 2009, San Diego, CA, October 11-14, 2009. INFORMS 2009, October 2009. , Oct-2009
Articles in Peer-reviewed Journals	Chiam TC, Yih Y, Mitchell CA. "Control policies for a water-treatment system using the Markov Decision Process. Part 1: Mathematical formulation." Habitation – An International Journal for Human Support Research. 2009 Nov;12(1):21-6. , Nov-2009
Articles in Peer-reviewed Journals	Chiam TC, Yih Y, Mitchell CA. "Control policies for a water-treatment system using the Markov Decision Process. Part 2: Simulation and analysis." Habitation – An International Journal for Human Support Research. 2009 Nov;12(1):27-32. , Nov-2009
Articles in Peer-reviewed Journals	Chiam TC, Yih Y. "Controlled Random Walk Approach to State Space Exploration for a Markov Decision Process. (Submitted)." Submitted to Computers in Industrial Engineering, November 2009. , Nov-2009
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Articles in Peer-reviewed Journals	Chiam TC, Yih Y. "A Rule-based system-approach to system health improvement in a NASA Advanced Life Support System. " In Progress, November 2009. , Nov-2009
Articles in Peer-reviewed Journals	Chiam TC, Yih Y, Mitchell CA. "Development of theoretical approaches to the control of a water-treatment system." Advances in Space Research. Article in Press, Accepted Manuscript, Available online 25 August 2010. http://dx.doi.org/10.1016/j.asr.2010.08.022 , Aug-2010
Dissertations and Theses	Chiam TC. "Control Strategy for Dynamic Sustainable Systems." Dissertation, Purdue University, Degree awarded May 2010. , May-2010